Step 2. Look for multiples common to both lists. If there are no common multiples in the lists, write out additional multiples for each number.
Step 3. Look for the smallest number that is common to both lists.
Step 4. This number is the LCM.

- **Find the LCM using the prime factors method.**
Step 1. Find the prime factorization of each number.
Step 2. Write each number as a product of primes, matching primes vertically when possible.
Step 3. Bring down the primes in each column.
Step 4. Multiply the factors to get the LCM.

Exercises

Review Exercises

Use the Language of Algebra

Use Variables and Algebraic Symbols

In the following exercises, translate from algebra to English.

317. $3 \cdot 8$

318. $12 - x$

319. $24 \div 6$

320. $9 + 2a$

321. $50 \geq 47$

322. $3y < 15$

323. $n + 4 = 13$

324. $32 - k = 7$

Identify Expressions and Equations

In the following exercises, determine if each is an expression or equation.

325. $5 + u = 84$

326. $36 - 6s$

327. $4y - 11$

328. $10x = 120$

Simplify Expressions with Exponents

In the following exercises, write in exponential form.

329. $2 \cdot 2 \cdot 2$

330. $a \cdot a \cdot a \cdot a \cdot a$

331. $x \cdot x \cdot x \cdot x \cdot x \cdot x$

332. $10 \cdot 10 \cdot 10$

In the following exercises, write in expanded form.

333. 8^4

334. 3^6

335. y^5

336. n^4

In the following exercises, simplify each expression.

337. 3^4

338. 10^6

339. 2^7

340. 4^3

Simplify Expressions Using the Order of Operations

In the following exercises, simplify.

341. $10 + 2 \cdot 5$

342. $(10 + 2) \cdot 5$

343. $(30 + 6) \div 2$

344. $30 + 6 \div 2$

345. $7^2 + 5^2$

346. $(7 + 5)^2$

347. $4 + 3(10 - 1)$

348. $(4 + 3)(10 - 1)$

Evaluate, Simplify, and Translate Expressions

Evaluate an Expression

In the following exercises, evaluate the following expressions.

349. $9x - 5$ when $x = 7$

350. y^3 when $y = 5$

351. $3a - 4b$ when $a = 10, b = 1$

352. bh when $b = 7, h = 8$

Identify Terms, Coefficients and Like Terms

In the following exercises, identify the terms in each expression.

353. $12n^2 + 3n + 1$

354. $4x^3 + 11x + 3$

In the following exercises, identify the coefficient of each term.

355. $6y$

356. $13x^2$

In the following exercises, identify the like terms.

357. $5x^2, 3, 5y^2, 3x, x, 4$

358. $8, 8r^2, 8r, 3r, r^2, 3s$

Simplify Expressions by Combining Like Terms

In the following exercises, simplify the following expressions by combining like terms.

359. $15a + 9a$

360. $12y + 3y + y$

361. $4x + 7x + 3x$

362. $6 + 5c + 3$

363. $8n + 2 + 4n + 9$

364. $19p + 5 + 4p - 1 + 3p$

365. $7y^2 + 2y + 11 + 3y^2 - 8$

366. $13x^2 - x + 6 + 5x^2 + 9x$

Translate English Phrases to Algebraic Expressions

In the following exercises, translate the following phrases into algebraic expressions.

367. the difference of x and 6

368. the sum of 10 and twice a

369. the product of $3n$ and 9

370. the quotient of s and 4

371. 5 times the sum of y and 1

372. 10 less than the product of 5 and z

373. Jack bought a sandwich and a coffee. The cost of the sandwich was \$3 more than the cost of the coffee. Call the cost of the coffee c. Write an expression for the cost of the sandwich.

374. The number of poetry books on Brianna's bookshelf is 5 less than twice the number of novels. Call the number of novels n. Write an expression for the number of poetry books.

Solve Equations Using the Subtraction and Addition Properties of Equality

Determine Whether a Number is a Solution of an Equation

In the following exercises, determine whether each number is a solution to the equation.

375. $y + 16 = 40$
ⓐ 24 ⓑ 56

376. $d - 6 = 21$
ⓐ 15 ⓑ 27

377. $4n + 12 = 36$
ⓐ 6 ⓑ 12

378. $20q - 10 = 70$
ⓐ 3 ⓑ 4

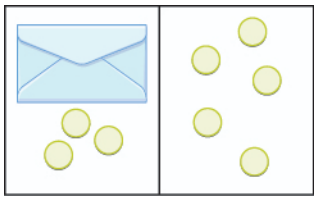
379. $15x - 5 = 10x + 45$
ⓐ 2 ⓑ 10

380. $22p - 6 = 18p + 86$
ⓐ 4 ⓑ 23

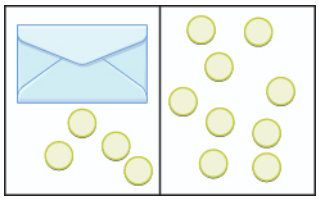
Model the Subtraction Property of Equality

In the following exercises, write the equation modeled by the envelopes and counters and then solve the equation using the subtraction property of equality.

381.



382.



Solve Equations using the Subtraction Property of Equality

In the following exercises, solve each equation using the subtraction property of equality.

383. $c + 8 = 14$

384. $v + 8 = 150$

385. $23 = x + 12$

386. $376 = n + 265$

Solve Equations using the Addition Property of Equality

In the following exercises, solve each equation using the addition property of equality.

387. $y - 7 = 16$

388. $k - 42 = 113$

389. $19 = p - 15$

390. $501 = u - 399$

Translate English Sentences to Algebraic Equations

In the following exercises, translate each English sentence into an algebraic equation.

391. The sum of 7 and 33 is equal to 40.

392. The difference of 15 and 3 is equal to 12.

393. The product of 4 and 8 is equal to 32.

394. The quotient of 63 and 9 is equal to 7.

395. Twice the difference of n and 3 gives 76.

396. The sum of five times y and 4 is 89.

Translate to an Equation and Solve

In the following exercises, translate each English sentence into an algebraic equation and then solve it.

397. Eight more than x is equal to 35.

398. 21 less than a is 11.

399. The difference of q and 18 is 57.

400. The sum of m and 125 is 240.

Mixed Practice

In the following exercises, solve each equation.

401. $h - 15 = 27$

402. $k - 11 = 34$

403. $z + 52 = 85$

404. $x + 93 = 114$

405. $27 = q + 19$

406. $38 = p + 19$

407. $31 = v - 25$

408. $38 = u - 16$

Find Multiples and Factors

Identify Multiples of Numbers

In the following exercises, list all the multiples less than 50 for each of the following.

409. 3

410. 2

411. 8

412. 10

Use Common Divisibility Tests

In the following exercises, using the divisibility tests, determine whether each number is divisible by 2, by 3, by 5, by 6, and by 10.

413. 96

414. 250

415. 420

416. 625

Find All the Factors of a Number

In the following exercises, find all the factors of each number.

417. 30

418. 70

419. 180

420. 378

Identify Prime and Composite Numbers

In the following exercises, identify each number as prime or composite.

421. 19

422. 51

423. 121

424. 219

Prime Factorization and the Least Common Multiple

Find the Prime Factorization of a Composite Number

In the following exercises, find the prime factorization of each number.

425. 84

426. 165

427. 350

428. 572

Find the Least Common Multiple of Two Numbers

In the following exercises, find the least common multiple of each pair of numbers.

429. 9, 15

430. 12, 20

431. 25, 35

432. 18, 40

Everyday Math

433. Describe how you have used two topics from The Language of Algebra chapter in your life outside of your math class during the past month.

Practice Test

In the following exercises, translate from an algebraic equation to English phrases.

434. $6 \cdot 4$

435. $15 - x$

In the following exercises, identify each as an expression or equation.

436. $5 \cdot 8 + 10$

437. $x + 6 = 9$

438. $3 \cdot 11 = 33$

439. (a) Write $n \cdot n \cdot n \cdot n \cdot n \cdot n$ in exponential form.
(b) Write 3^5 in expanded form and then simplify.

In the following exercises, simplify, using the order of operations.

440. $4 + 3 \cdot 5$

441. $(8 + 1) \cdot 4$

442. $1 + 6(3 - 1)$

443. $(8 + 4) \div 3 + 1$

444. $(1 + 4)^2$

445. $5[2 + 7(9 - 8)]$

In the following exercises, evaluate each expression.

446. $8x - 3$ when $x = 4$

447. y^3 when $y = 5$

448. $6a - 2b$ when $a = 5, b = 7$

449. hw when $h = 12, w = 3$

450. Simplify by combining like terms.

ⓐ $6x + 8x$

ⓑ $9m + 10 + m + 3$

In the following exercises, translate each phrase into an algebraic expression.

451. 5 more than x

452. the quotient of 12 and y

453. three times the difference of a and b

454. Caroline has 3 fewer earrings on her left ear than on her right ear. Call the number of earrings on her right ear, r. Write an expression for the number of earrings on her left ear.

In the following exercises, solve each equation.

455. $n - 6 = 25$

456. $x + 58 = 71$

In the following exercises, translate each English sentence into an algebraic equation and then solve it.

457. 15 less than y is 32.

458. the sum of a and 129 is 164.

459. List all the multiples of 4 that are less than 50.

460. Find all the factors of 90.

461. Find the prime factorization of 1080.

462. Find the LCM (Least Common Multiple) of 24 and 40.